EXERCISE 1
C# vs Java Naming and Formatting Conventions

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1. Naming Conventions

Item	C#	Java	Description
Class / Type	PascalCase	PascalCase	Both start each word with a capital letter.
Method	PascalCase	camelCase	C# starts with uppercase; Java starts with lowercase and usually uses a verb.
Variable / Parameter	camelCase	camelCase	Both normally start with lowercase.
Constant	PascalCase	UPPER_CASE	Java uses uppercase with underscores; C# avoids screaming caps.
Interface	I + PascalCase	PascalCase	C# commonly adds I, e.g. IShape. Java has no required I-prefix.
Private member	_name (allowed/common)	name_	The cited C# guide allows underscore prefix; GeoSoft Java uses underscore suffix.
Package / Namespace	PascalCase parts	lowercase	C# namespaces are structured; Java package names are lowercase.

2. Formatting Conventions

Item	C#	Java	Description
Curly braces	Opening brace is normally on the next line.	Usually opening brace is on the same line.	Both use braces to show blocks; the main difference is placement.
Indentation	Use consistent indentation.	GeoSoft recommends 2 spaces.	Indentation makes nested code easier to read.
Line length	No fixed limit in the cited C# standards.	GeoSoft recommends 80 columns.	Long Java lines should be split clearly.
Blank lines	Use whitespace consistently.	Use blank lines to separate logical sections.	Both use whitespace to improve readability.

File name	Usually matches the main class.	.java file normally matches the public class.	Example: Customer.cs / Customer.java.
Member organization	Member variables are placed at the top of a class.	Variables, constructors and methods follow a predictable order.	The goal is to make code easy to find.

3. Simple Examples

C#:

```
class Student
{
    public void CalculateScore()
    {
        int totalScore = 90;
    }
}
```

Java:

```
class Student {
  public void calculateScore() {
    int totalScore = 90;
  }
}
```

4. Conclusion

C# and Java are similar in class and variable naming, but they have some important differences. C# uses PascalCase for methods, while Java uses camelCase. Java constants normally use UPPER_CASE_WITH_UNDERSCORES, while C# does not. Formatting is also different mainly in brace style, indentation, and line length. The main purpose of both conventions is to make code clear, consistent, and easy to maintain.

5. References

1. C# Coding Guidelines – https://csharpcodingguidelines.com/
2. Dofactory – C# Coding Standards – https://www.dofactory.com/csharp-coding-standards
3. Oracle – Code Conventions for the Java Programming Language – https://java.sun.com/docs/codeconv/html/CodeConvTOC.doc.html
4. GeoSoft – Java Programming Style Guidelines – https://geosoft.no/javastyle.html#Naming_Conventions